

Early Identification of Traumatic Pneumothorax in a Pediatric Patient using PALS Protocols

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1. Case Summary & Clinical Presentation

A 7-year-old child was presented to the Emergency Department via ambulance following a Motor Vehicle Accident (MVA). The mother, who was the driver, was also admitted. On initial "Across-the-Room" observation (the **Pediatric Assessment Triangle**), the child appeared stable: he was conscious, breathing without significant distress, and had normal skin color.

The Systematic Assessment

Despite the apparent stability, I applied the **PALS Primary Assessment (ABCDE)**:

- **Airway:** Patent; the child was speaking.
- **Breathing:** Upon closer inspection and palpation, I noted a subtle but definitive **tracheal deviation** away from the right side. Auscultation revealed diminished breath sounds on the right.
- **Circulation:** Heart rate was slightly tachycardic for age (125 bpm), though BP remained within normal limits.

The Intervention

Recognizing the tracheal shift as a sign of an impending Tension Pneumothorax:

1. **Imaging:** I ordered an immediate STAT portable Chest X-ray while preparing for decompression.
2. **Diagnosis:** The X-ray confirmed a significant **right-sided pneumothorax**.
3. **Action:** Under PALS guidelines for trauma, I led the team in stabilized oxygenation and coordinated with the surgical team for the insertion of a chest tube (thoracostomy).

2. Measuring the Outcome

The "Measuring Outcomes" focus here is on preventing obstructive shock. Identifying the tracheal shift *before* the patient became hypotensive significantly altered the outcome.

Table 1: Clinical Data and Intervention Impact

Clinical Parameter	Initial Assessment (Pre-Intervention)	Post-Intervention (Chest Tube)	Clinical Significance
Tracheal Position	Deviated to the Left.	Midline.	Normalization of mediastinal structures.

Air Entry	Diminished on the Right.	Equal bilaterally.	Improved lung expansion and ventilation.
Heart Rate	125 bpm (Tachycardia).	95 bpm (Normal for age).	Resolution of compensatory tachycardia.
Oxygen Saturation	94% on Room Air.	99% on Room Air.	Optimized oxygenation.

3. Clinical Reasoning & Protocol Adherence

This case demonstrates the efficacy of the **PALS Systematic Approach Algorithm (Evaluate-Identify-Intervene)**.

- **Evaluate:** The tracheal deviation was a specific finding identified during the "Evaluate" phase that did not match the "stable" appearance in the initial Assessment Triangle.
- **Identify:** I identified this as a **"Life-Threatening Breathing Problem"** (Tension Pneumothorax physiology).
- **Intervene:** Early intervention prevented the shift from obstructing venous return to the heart, which would have led to cardiac arrest.

4. Reflection on Practice

What Went Well

The use of the PALS Systematic Approach ensured that the "stable" appearance of the child did not lead to complacency. Pediatric patients compensate extremely well until they reach a "cliff" of sudden collapse; catching the tracheal deviation allowed us to intervene while the child still had physiological reserves.

Areas for Improvement

In the moment, there was a slight delay in the arrival of the portable X-ray. In a true "tension" scenario (hemodynamic instability), I reflect that I should have been prepared to perform a **needle decompression** immediately without waiting for imaging.

Action Plan for Future Practice

- **Skill Maintenance:** I will attend a simulation session on pediatric needle decompression and chest tube thoracostomy to ensure technical proficiency under pressure.

5. Conclusion

The application of PALS training in this case resulted in the early diagnosis of a traumatic pneumothorax. By measuring the normalization of the patient's heart rate and the restoration of midline tracheal anatomy, I can objectively state that the systematic approach prevented a high-morbidity event (obstructive shock).